

Effect of Cu content on central-segregation composition and mechanical properties of Al–Mg–Si alloys produced by twin-roll casting

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ABSTRACT

Dissolving the inherent central segregation is critical to improve the performance of twin-roll casting (TRC) aluminum strips. In this study, we investigated the effects of Cu content (0.8, 1.2, and 1.6 wt%) on the central-segregation composition and mechanical properties of TRC Al–0.8Mg–1.0Si alloys. Cu promoted the segregation-phase transformation from the higher-melting-temperature Mg₂Si phase to the lower-melting-temperature Cu-containing phases (Q-AlCuMgSi and θ -Al₂Cu), resulting in an easier dissolution of central segregation. Therefore, the segregation of 0.8 and 1.2 wt% Cu alloys was almost dissolved following cold rolling and solution treatment compared to Cu-free alloys. However, after a higher content of Cu (1.6 wt%) was added, the segregation became too severe to dissolve. After modifying with 1.2 wt% Cu, the yield strength (YS) of the TRC alloy increased from ~306 to 335 MPa, and the elongation was increased from ~13 to 17%. The contribution of solution strengthening and increasing volume fraction of precipitates results in the strengthening effect of Cu. Meanwhile, the dissolution of segregated phases is also advantageous for solution and precipitation strengthening. This study could lead to the development of a simple method to dissolve TRC central segregation and produce TRC Al–Mg–Si alloys with excellent performance.

1. Introduction

Age-hardened Al–Mg–Si alloys have recently been used in automobiles and aerospace due to their low density and high strength [1,2]. Traditional production technologies, such as direct-chill casting (DC), limited their widespread application due to their high cost [3]. Twin-roll casting (TRC) is a near-net-shape forming technology that requires less equipment investment, energy consumption, and space occupancy when compared to DC [4,5]. Moreover, Cu–Be rolls provide better thermal conductivity in the recently developed twin-roll casters. The cooling rate reaches sub-rapid solidification (SRS) condition (about 10^2 – 10^3 K/s [6]), which facilitates the production of TRC strips with finer grains and intermetallic compounds.

However, TRC exhibits inherent flaws, such as central segregation and insufficient mechanical properties, which must be addressed [7,8]. The formation of central segregation is linked to the accumulation of insoluble secondary phases after solidification. Therefore, controlling the solidification process is a possible solution to restrain the central

segregation. Electromagnetic fields [9,10] and grain refiners [11–14] have been reported to promote the nucleation and then effectively reduce the as-cast central segregation. According to Chen et al. [9], the electromagnetic oscillation field in the TRC process developed a fine nondendritic microstructure and increased the number of nucleation cores, resulting in fewer segregation defects in commercial AA6022 alloy. According to Liu et al. [11], TiC nanoparticles (0.5 wt%) in Al–5Cu alloys stimulated the formation of equiaxed grains, and the well-connected liquid flow channels between the grains prevented the accumulation of Cu-rich melt. Moreover, Fu et al. [15] reported that reducing the rolling speed (to 9.83 mm/s) to control the solidification process of AA5059 alloy strips could also eliminate the central segregation and improve the hardness uniformity. However, the complexity of melting or solidification processes makes them difficult for manufacturers to accurately control. Therefore, developing a more straightforward and effective approach to dissolve TRC central segregation during subsequent heat treatment is a more feasible method.

Several studies have reported the effect of heat treatment on TRC

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central segregation. According to Zhang et al. [16], the combination of controlled rolling and solid solution treatment effectively dissolved the segregated phases of the TRC Mg–8Al–2Sn–1Zn (ATZ821) alloy. Liu et al. [17] transformed π -AlFeMgSi into α -Al(FeMn)Si phase by adding 0.3 wt% Mn to the TRC Al–0.5Mg–0.7Si alloy, resulting in a more uniform heat-treated microstructure. However, early experimental studies rarely considered the effect of central-segregation composition. The addition of Cu to Al–Mg–Si alloys, according to Chakrabarti et al. [18], can cause significant changes in the second phase, primarily the formation of Cu-containing phases (Q -AlCuMgSi and θ -Al₂Cu) and the reduction or even disappearance of Mg₂Si. Although Cu-containing phases have lower melting temperatures than Mg₂Si, the effect of these second-phase variations on central-segregation composition has not been studied. Thus, modifying the second-phase composition appears to be a potential method to dissolve central segregation. Cu is a traditional strengthening element that has been widely used in commercial 6xxx series alloys (for example, in 6016 and 6111 alloys [3]), in addition to influencing the second phase. The extensive application of Cu suggests that it affects not only the composition of the central segregation but also the mechanical properties of the TRC strips. Furthermore, trace element alloying is a widely used method [19–24] that meets our previous criteria for simplicity and effectiveness. Cu may thus play a critical role in enhancing the quality of TRC Al–Mg–Si strips.

The effect of Cu content on central-segregation composition and mechanical properties of SRS-TRC Al–Mg–Si strips was investigated in this study. The results are expected to be a valuable guide for dissolving TRC central segregation and producing high-quality TRC Al–Mg–Si alloys.

2. Experimental

In this study, an Al–Mg–Si–(Cu) alloy was used as the experimental material. The chemical compositions of the four Al–Mg–Si alloys with varying Cu contents and a similar Mg/Si ratio of ~ 0.95 are listed in Table 1. In addition to the increase in Cu, the Mg and Si contents in the 1.6Cu alloy were deliberately increased to enable similar second-phase types as the 1.2Cu alloy.

As raw materials for Al–Mg–Si–(Cu) alloys, commercial high-purity Al (99.9%), high-purity Mg (99.9%), and master alloys (Al–24.4 wt% Si, Al–50 wt% Cu) were used. Fe is an impurity derived from raw materials. After melting high-purity Al in the muffle furnace, the master alloys and high-purity Mg were immersed in the liquid, and a heating temperature of 720 °C was maintained until all the materials were completely melted. The melt was poured into twin rollers after the temperature was reduced to 700 °C for the TRC experiment. The diagram of the industrial twin-roll casters is shown in Fig. 1a, and the TRC process route used in this study is shown in Fig. 1b.

The TRC was performed on an inclined twin-roll caster with water-cooled Cu–Be rolls, and the connecting line between the centers of the two rolls was inclined at an angle of 20° to the vertical (Fig. 1b). The vacuum degree in the melting chamber was ~ 0.7 Pa, and the diameter of the rolls was ~ 230 mm. The rolling speed is ~ 7 m/min, and the as-received TRC strips were ~ 4 mm thick and ~ 130 mm wide. The TRC strips were first homogenized for 2 h at 535 °C. The strips were cold-rolled to a final thickness of ~ 1 mm after being air-cooled to room temperature (25 °C). The cold-rolled strips were solution treated for 10

min at 535 °C, followed by water quenching, and then aged artificially at 175 °C.

The samples for microstructure analysis were cut along the cross-section of the strips. Following grinding and polishing, the cross-sectional surfaces of samples were anodized for polarized optical microscopy (OM, Zeiss Axio Imager A2M, Germany) using Barker's reagent (1.8% fluorescent acid in water) at 20–45 V DC for up to 2 min. Image-Pro Plus software was used to analyze OM images to determine secondary dendrite arm spacing (SDAS). The actual composition of the TRC strips was determined using an optical spectrum analyzer (ARL 4460, Thermo Fisher, Switzerland). Scanning electron microscopy (SEM, Zeiss Sigma 500, backscattered electron mode, BSE, Germany) and energy dispersive spectroscopy (EDS, INCA-X-Max, England) were used to characterize the morphology and distribution of the central segregations. The samples for electron backscatter diffraction (EBSD) were electropolished for 30 s at 25 V with a perchloric acid ethanol solution (10 vol% perchloric acid and 90 vol% ethanol), and the resulting EBSD images were analyzed using AZtec software. To characterize precipitates, a transmission electron microscope and high-resolution transmission electron microscopy (HRTEM, FEI Talos F200X, USA) were used at 200 kV, and the samples were prepared by twin-jet electron polishing in a solution of 30% nitric acid and 70% methanol at about -30 °C.

The microhardness value was measured using a Vickers microhardness tester (HVS-1000A, HuaYin, China) with a load of 200 g and a dwell time of 15 s. The average microhardness value was calculated using at least 10 micro indentations for each sample. Tensile tests performed on a material testing machine (AGS-X, SHIMADZU, Japan) at a strain rate of ~ 0.6 mm/min, with at least three samples prepared along the rolling direction for each strip.

3. Results

3.1. As-cast microstructures

Fig. 2 depicts polarized OM images of as-cast TRC alloys with varying Cu contents. The microstructures exhibited a typical multiplex structure distribution, with segregation bands near the top of the strips. In the 0Cu sample (Fig. 2a), the SDAS of the segregation band zone (indicated by dashed white lines) is ~ 20.4 μ m. As the Cu content increases (Fig. 2b–d), the SDAS refines to ~ 16.0 , 14.7, and 14.4 μ m, respectively. These findings suggest that the addition of Cu decreased the SDAS of the segregation bands.

Fig. 3a–f shows SEM–BSE images of as-cast TRC alloys with varying Cu contents, demonstrating the central segregation and dendritic intermetallic compounds. Segregation is occasionally observed along the segregation band in 0Cu alloy (Fig. 3a). The central segregation is not reduced by the addition of 0.8 wt% Cu (Fig. 3d). According to the EDS results, the segregation in 0Cu alloy marked in Fig. 3b consists of Mg₂Si (point A: Al: 60.7%, Mg: 24.6%, Si: 14.7%, at.%) and AlFeMgSi (average of points B and C: Al: 77.9%, Mg: 6.1%, Si: 12.6%, Fe: 3.4%, at.%) phases. Moreover, the second phases outside the segregation band in 0Cu alloy, marked in Fig. 3c, are Mg₂Si (point D: Al: 51.5%, Mg: 26.1%, Si: 21.6%, Fe: 0.8%, at.%) and AlFeSi (point E: Al: 84.6%, Mg: 0.5%, Si: 9.9%, Fe: 5.0%, at.%) phases. With the addition of 0.8 wt% Cu, the segregation in 0.8Cu alloy marked in Fig. 3e can be considered as a mixture of Q -AlCuMgSi, AlFeMgSi, and θ -Al₂Cu phases (average of points F and G: Al: 84.7%, Mg: 6.1%, Si: 6.8%, Cu: 1.9%, Fe: 0.5%, at.%). Furthermore, the second phases outside the segregation band in 0.8Cu alloy, marked in Fig. 3f, can be considered as a mixture of Q -AlCuMgSi and θ -Al₂Cu phases (average of points H and I: Al: 90.4%, Mg: 1.7%, Si: 5.2%, Cu: 2.7%, at.%). These results indicate that Cu changes the Mg₂Si phase into the Q -AlCuMgSi and θ -Al₂Cu phases. Because the segregation of 1.2Cu (Figs. 3g) and 1.6Cu (Fig. 3h) alloys has the same morphology and composition as the 0.8Cu alloy, no additional details will be shown in this study.

Table 1
Chemical compositions of the four Al–Mg–Si alloys in this study (wt.%).

	Mg (wt. %)	Si (wt. %)	Cu (wt. %)	Fe (wt. %)	Al	Mg/Si (at. %)
0Cu	0.86	1.09	–	0.10	Bal.	0.92
0.8Cu	0.85	1.01	0.77	0.10	Bal.	0.97
1.2Cu	0.85	1.04	1.22	0.09	Bal.	0.95
1.6Cu	0.99	1.21	1.65	0.12	Bal.	0.95

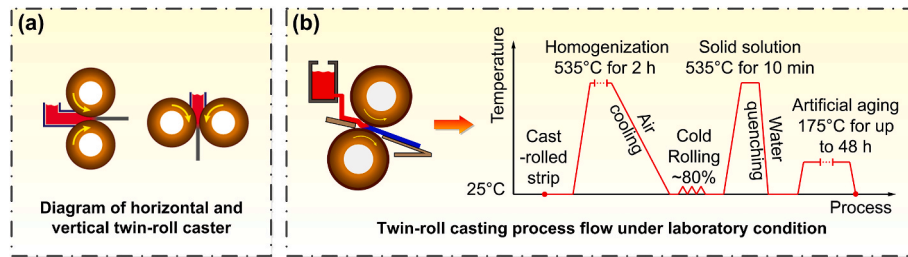


Fig. 1. Schematic diagram of TRC production, (a) the diagram of two kinds of industrial twin-roll casters, (b) the TRC process route of this study.

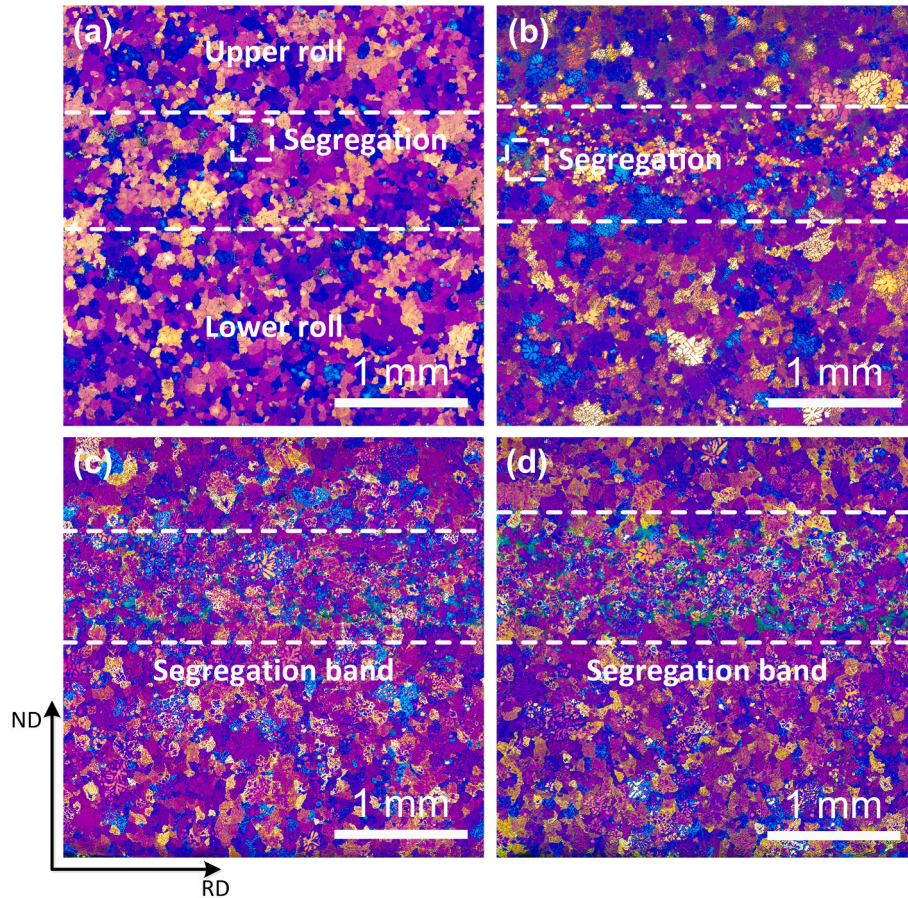


Fig. 2. As-cast microstructures of Al-Mg-Si(-Cu) alloys (a) 0Cu alloy, (b) 0.8Cu alloy, (c) 1.2Cu alloy, and (d) 1.6Cu alloy.

3.2. Microstructures after cold rolling and solution treatment

SEM-BSE images of TRC alloys with varying Cu contents following cold rolling and solution treatment are shown in Fig. 4a–f. The central segregation in 0Cu alloy was detected as block-like AlFeSi (point A in Fig. 4a: Al: 87.6%, Mg: 0.6%, Si: 7.4%, Fe: 4.5%, at.%) and dark residual Mg₂Si (point B in Fig. 4a: Al: 63.2%, Mg: 18.6%, Si: 18.2%, at.%) phases, according to the EDS results. With the addition of 0.8 wt% Cu, only a few finely dispersed residual Mg₂Si and AlFeSi (point C in Fig. 4b: Al: 90.5%, Mg: 0.5%, Si: 5.8%, Fe: 3.2% at.%) phases are discovered. Similarly, despite the small number of needle-like AlFeSi and residual Mg₂Si phases (Fig. 4c and e), the central segregation in the 1.2Cu alloy was nearly dissolved. It is worth noting that Cu-containing phases are absent from 0.8Cu and 1.2Cu alloys. Moreover, the segregation (shown in Fig. 4d and f as a mixture of Mg₂Si, AlFeMgSi, Q-AlCuMgSi, and θ -Al₂Cu phases) did not dissolve in the 1.6Cu alloy. In summary, it is apparent from Fig. 4 that following cold rolling and solution treatment, the segregation of 0.8Cu and 1.2Cu is almost completely dissolved, whereas

severe segregation remains in the 1.6Cu alloy.

The EBSD inverse pole figures (IPF) and corresponding grain size statistical diagrams of TRC alloys with different Cu contents after cold rolling and solution treatment are shown in Fig. 5a–h. The average grain sizes of the 0Cu–1.6Cu alloys are ~22.9, 26.1, 31.4, and 27.1 μm , respectively. According to Shi et al. [25], the recrystallized grain size was related to the volume fraction of the second phases, which retarded the migration of grain boundary due to their pinning effect. The dissolution of segregation phases in 0.8Cu and 1.2Cu alloys reduced the pinning force, leading to a gradual increase in grain size. However, the grain size decreased as the segregation of the 1.6Cu alloy became severe. All four samples had similar recrystallized microstructures, indicating that Cu cannot suppress the recrystallization in quenched alloys.

3.3. Mechanical properties

Fig. 6 shows the hardness curves of the alloys with varying Cu contents aged at 175 °C. It can be seen that the 0Cu alloy reached a peak-

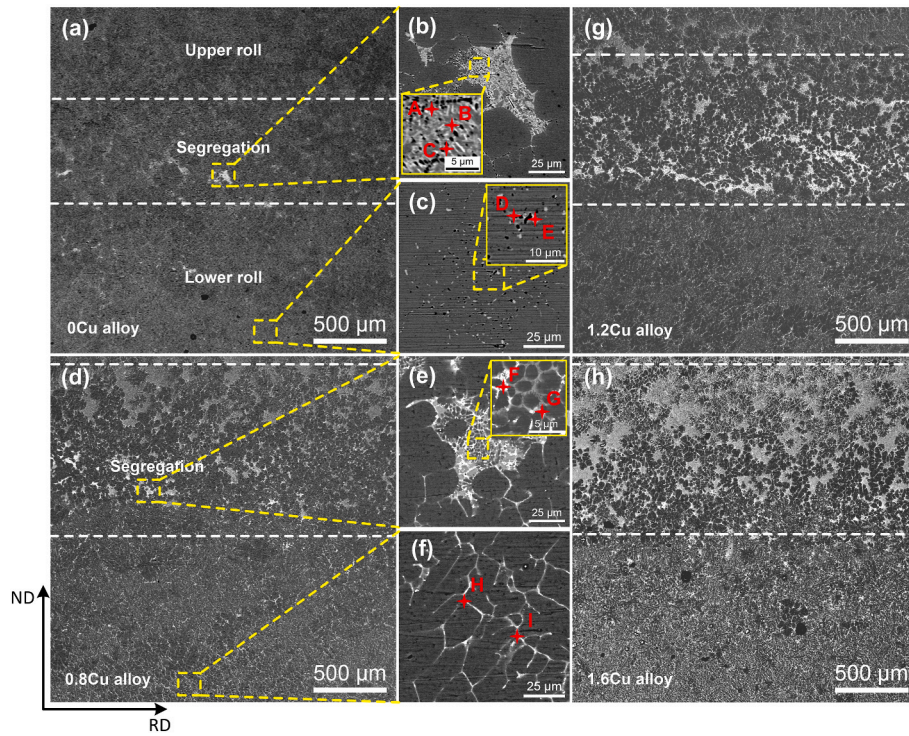


Fig. 3. SEM-BSE images of the four as-cast Al-Mg-Si alloys: (a)–(c) 0Cu alloy; (d)–(f) 0.8Cu alloy; (g) 1.2Cu alloy; (h) 1.6Cu alloy.

aging (PA) hardness value of ~ 134.9 HV following 12 h of aging. After alloying with 0.8, 1.2, and 1.6 wt% Cu, the hardness values improve significantly to ~ 148.1 , 153.3, and 152.5 HV after 12, 12, and 8 h, respectively. The addition of Cu effectively increased the PA hardness of the TRC alloys.

The engineering stress–strain curves of the four alloys in as-quenched (AQ) and PA states are shown in Fig. 7, and the specific values of YS, ultimate tensile strength (UTS), and elongation to fracture (El) are listed in Table 2. The YS and El of 0Cu alloy at AQ state are ~ 71 MPa and $\sim 30\%$, respectively (Fig. 7a). Cu addition slightly increases the YS and El to a maximum of ~ 72 MPa (0.8Cu alloy) and $\sim 32\%$ (1.2Cu and 1.6Cu alloys), indicating that Cu addition has not significantly reduced the cold workability of TRC alloys. For PA alloys (Fig. 7b), the YS and El increase with the addition of 1.2 wt% Cu from ~ 306 MPa to $\sim 13\%$ to ~ 335 MPa and $\sim 17\%$, respectively. When the Cu content was increased to 1.6 wt%, the El decreased to $\sim 16\%$ and the YS increased by only ~ 1 MPa. In other words, once the Cu content exceeds 1.2 wt%, a further increase in Cu content will not significantly improve the mechanical properties of TRC alloys. According to the as-cast and AQ microstructures shown in Figs. 3 and 4, this study considers the alloy with 1.2 wt% Cu to be the best Cu content in the four TRC Al-Mg-Si alloys.

The bright-field images of the PA alloys viewed parallel to the $\langle 001 \rangle_{\text{Al}}$ zone axis are shown in Fig. 8a–d, and the HRTEM images and corresponding fast Fourier transform (FFT) patterns of the two representative precipitates are shown in Fig. 9a–d. Fig. 8 shows two types of precipitates: one with a nearly circular cross-section and a lath-shaped morphology (indicated by red dashed lines), and the other with a thin rectangular cross-section and a needle-shaped morphology (indicated by yellow dashed lines). The former precipitate exhibits a large cross-sectional area and a short length, which widely distributes in the four alloys (Fig. 8a–d), and has been identified as the β'' phase (with a monoclinic structure; the lattice parameters are $a = 15.16$ Å, $b = 4.05$ Å, $c = 6.74$ Å, $\beta = 105.3^\circ$) [6]. The latter precipitate appears in Cu-containing alloys and is recognized as a Q' -precursor, with a smaller cross-sectional area and a large length (Fig. 8b–d) [26,27]. The blue flat hexagonal atomic column in Fig. 9c represents a C phase feature, while

the red line represents a C unit cell (a typical precursor of Q' with a monoclinic structure; the lattice parameters are $a = 10.32$ Å, $b = 8.10$ Å, $c = 4.05$ Å, $\gamma = 100.9^\circ$) [28]. The QP lattice (yellow lines in Fig. 9c) is reflected by the QP spots (yellow circles in Fig. 9d).

A quantitative TEM analysis was performed in at least five fields per sample. The average length (L) was determined by counting ~ 300 to 400 precipitates per sample, and the average cross-sectional area (A) was determined by analyzing 50 HRTEM images. The number density of the observation area can be expressed as [29].

$$\rho = \frac{3N}{A_v(t+L)} \quad (1)$$

where N represents the number of precipitates counted over a view region area of A_v and t represents the thickness of the viewable region (estimated by convergent beam electron diffraction (CBED) to be approximately 222.1 nm). The volume fraction of the precipitates (f) can be calculated using the specific analytical method for thickness estimation [30] by multiplying the number density by the average length and cross-sectional area of the precipitates, and the results are shown in Table 3. Cu increased the precipitate length while decreasing the cross-sectional area and radius. Moreover, the number density and volume fraction of precipitates in the 0Cu alloy were $\sim 5.3 \times 10^{22}/\text{m}^{-3}$ and $\sim 6.8 \times 10^{-3}$, respectively, before alloying with 1.2 wt% Cu, and they increased to $\sim 7.5 \times 10^{22}/\text{m}^{-3}$ and $\sim 8.7 \times 10^{-3}$ after alloying. The number density barely changed after increasing the Cu content to 1.6 wt%, and the volume fraction improved by $\sim 1.3 \times 10^{-3}$ when compared to 1.2Cu alloy. In summary, Cu significantly improved the precipitate number density and volume fraction.

The YS and elongation of existing 6xxx alloys used for automobile outer panels in PA and paint-bake states are summarized in Fig. 10 [31–53]. As can be observed, the PA TRC alloys used in this study have an exceptional combination of medium YS and superior elongation. TRC alloys, in other words, demonstrate the advantages and potential of their mechanical properties over conventional alloys.

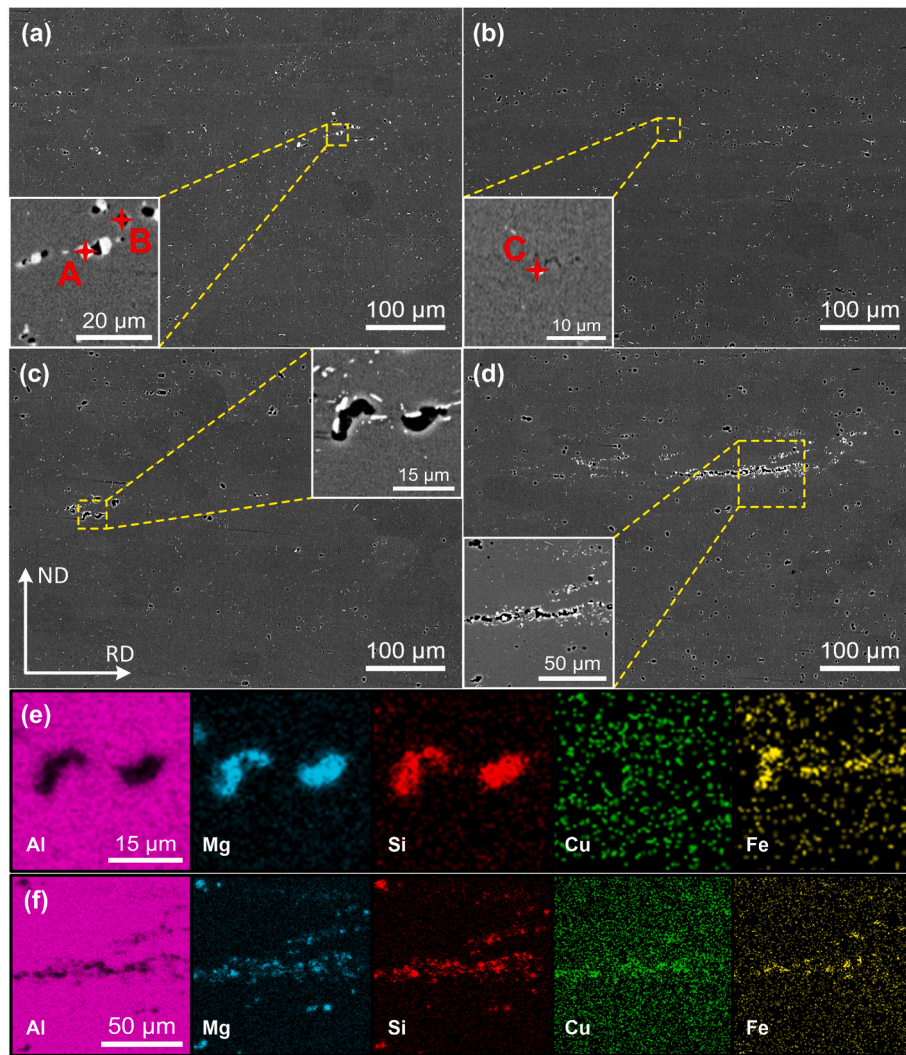


Fig. 4. SEM-BSE images of the four Al–Mg–Si strips after cold rolling and solution treatment: (a) 0Cu alloy, (b) 0.8Cu alloy, (c) 1.2Cu alloy, (d) 1.6Cu alloy; (e) and (f) are EDS element maps of the inset images in (c) and (d), respectively.

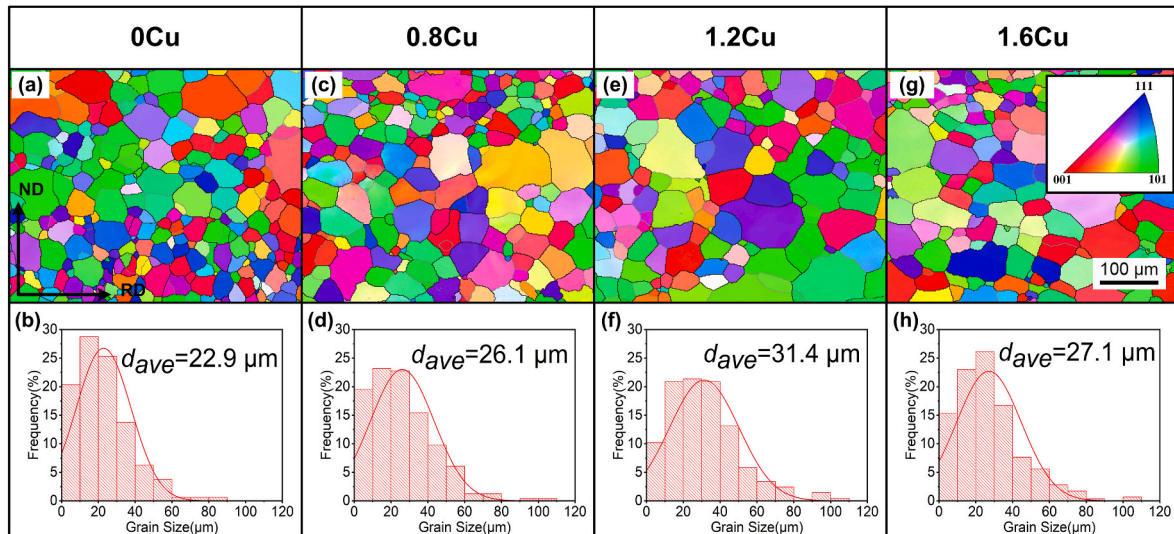


Fig. 5. EBSD grain structures and the corresponding grain size statistical diagrams of 0Cu–1.6Cu alloys after cold rolling and solid solution: (a)–(b) 0Cu alloy; (c)–(d): 0.8Cu alloy; (e)–(f) 1.2Cu alloy; (g)–(h) 1.6Cu alloy.

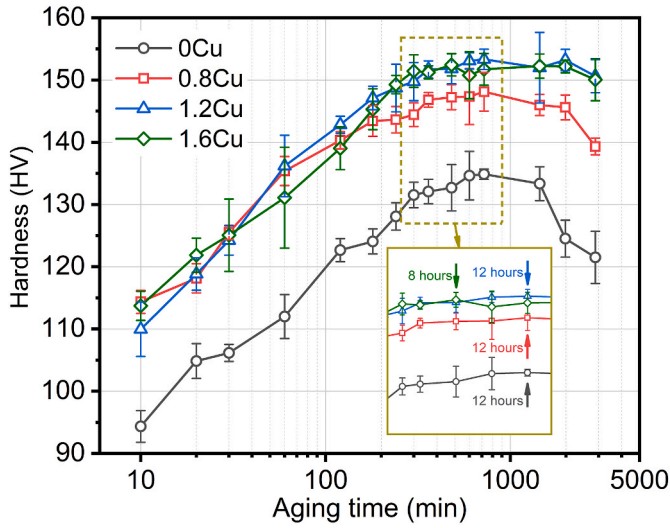


Fig. 6. The hardness curves of the four Al-Mg-Si alloys during artificial aging (AA).

4. Discussion

4.1. Relationship between segregation and solidification path

Pandat software was used to analyze the solidification paths of the four Al-Mg-Si alloys with varying Cu contents. The Scheil model was used to simulate phase diagrams, and the results are shown in Fig. 11. The second phase of 0Cu alloy consists of Mg_2Si , Si, β -AlFeSi, and $Al_8FeMg_3Si_6$ (dissolving temperatures: ~ 565 , 550, 600, and 565 $^{\circ}C$, respectively). Cu alloying produced two new phases: Q -AlCuMgSi and θ - Al_2Cu (dissolving temperatures: ~ 520 and 505 $^{\circ}C$, respectively). According to the calculation results, the α -AlFeSi phase fractions of all four alloys are quite low (as shown in Fig. 11a-d) and can therefore be reasonably ignored. Furthermore, as shown in Fig. 3e and f, $Al_8FeMg_3Si_6$ is discovered in the central segregation rather than β -AlFeSi or Mg_2Si which may be due to the special non-equilibrium solidification condition of SRS-TRC [45,54]. Mg_2Si , Si, and $Al_8FeMg_3Si_6$ phases were combined with Cu to form Q -AlCuMgSi phases, and excess Cu elements formed θ - Al_2Cu . During TRC, this phase transition causes most Mg_2Si phases in 0Cu alloy to transform into Q -AlCuMgSi phases in Cu-containing alloys, resulting in a change in second phase dissolution temperature (which is reduced by 45 $^{\circ}C$ from ~ 565 to 520 $^{\circ}C$). A lower dissolving temperature makes dissolving segregation easier. Moreover, Liu et al. [45] discovered that following cold rolling and solid solution, the $Al_8FeMg_3Si_6$ phase in the central segregation decomposed into

AlFeSi phases, explaining the presence of a large number of AlFeSi phases in Fig. 4a-c.

The calculated solidification paths are in general agreement with the results observed in Fig. 3. The EDS results in Section 3.1 also verify the solidification path calculations. These findings indicate that the addition of Cu significantly changes the composition of the central segregation from the higher-melting-temperature Mg_2Si phase to the lower-melting-temperature Cu-containing phases (Q -AlCuMgSi and θ - Al_2Cu), which reduces the difficulty of dissolving the central segregation. It thus makes the dissolving of central segregation by heat treatment possible. In addition, the compositional transformation and evolution process of the as-cast central segregation during cold-rolling and solution treatment are depicted schematically in Fig. 12. In the 0.8Cu and 1.2Cu alloys, a small number of residual Mg_2Si and fine AlFeSi phases were nearly dispersed along the segregation bands, significantly improving microstructural uniformity. The segregation of the 1.6Cu alloy, in contrast, was extremely severe to be eliminated, indicating that the 1.6Cu alloy was not suitable for TRC in this study.

4.2. Strengthening mechanism

Generally, PA Al-Mg-Si(-Cu) alloys are assumed to have a total strength that includes the inherent strength of the metal itself and the strength contributed by solute atoms and nano-precipitates. Based on the existing simulation models of the YS, the intensity contribution of each component is assumed to be a linear superposition [55,56]:

$$\sigma_y = \sigma_{gs} + \sigma_p + \sigma_{ss} \quad (4)$$

where σ_{gs} represents the effect of grain size, which follows the Hall-Petch relation [57].

$$\sigma_{gs} = \sigma_0 + kd^{-\frac{1}{2}} \quad (5)$$

Table 2

Mechanical properties of the four Al-Mg-Si alloys with different Cu contents at AQ and PA states.

Alloy	YS (MPa)		UTS (MPa)		EI (%)	
	AQ	PA	AQ	PA	AQ	PA
0Cu	70.9 $\pm$ 2.9	305.9 $\pm$ 2.1	181.8 $\pm$ 5.6	349.2 $\pm$ 1.8	30.4 $\pm$ 3.4	12.9 $\pm$ 1.2
0.8Cu	71.8 $\pm$ 1.6	325.4 $\pm$ 9.2	203.7 $\pm$ 3.5	386.7 $\pm$ 11.3	30.7 $\pm$ 2.6	16.7 $\pm$ 0.5
1.2Cu	71.2 $\pm$ 1.1	334.9 $\pm$ 2.6	216.0 $\pm$ 6.9	396.0 $\pm$ 4.4	31.9 $\pm$ 2.3	16.7 $\pm$ 2.6
1.6Cu	71.4 $\pm$ 3.7	336.3 $\pm$ 5.6	224.1 $\pm$ 1.9	414.8 $\pm$ 1.3	32.2 $\pm$ 0.9	16.2 $\pm$ 1.7

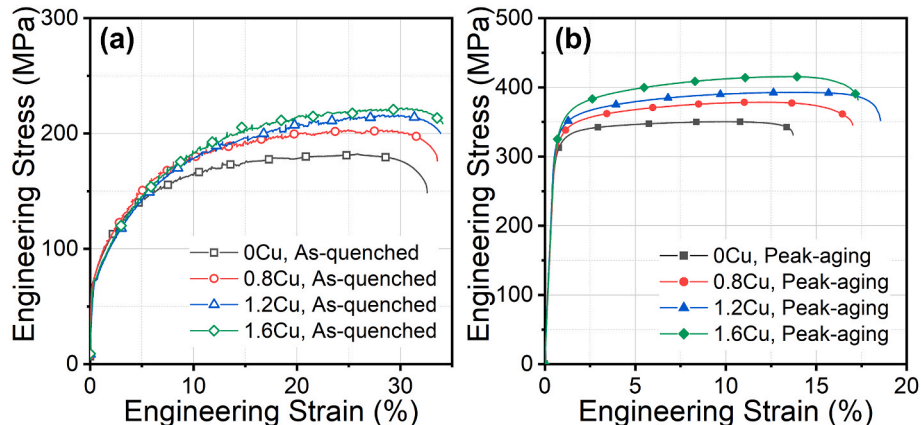


Fig. 7. Engineering stress-strain curves of the AQ and PA Al-Mg-Si alloys with different Cu contents: (a) as-quenched state, (b) peak-aging state.

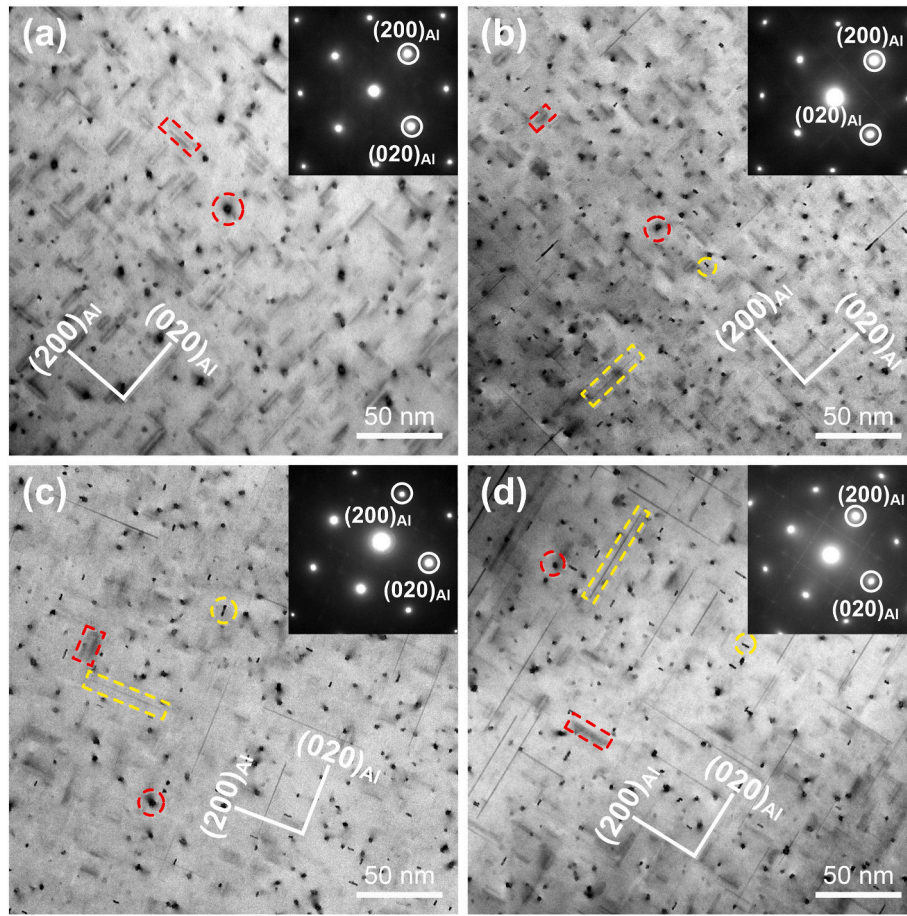


Fig. 8. TEM bright-field images and inset electron diffraction patterns of the four peak-aged alloys: (a) 0Cu alloy, (b) 0.8Cu alloy, (c) 1.2Cu alloy, and (d) 1.6Cu alloy.

where σ_0 represents the aluminum friction stress (~ 10 MPa [26]), k represents a constant commonly assumed to be $0.14 \text{ MPa m}^{1/2}$ in aluminum [58], and d represents the average grain boundary diameter, as mentioned in Section 4.2. Moreover, σ_p and σ_{ss} denote the YS contributed by nano-precipitates and by the solute atoms, respectively. The contribution of grain boundary strengthening is minor in 6xxx alloys [55]. Furthermore, the dislocation density in the recrystallized structure was not visible, and its effect on the strength was negligible [59]. The AQ alloys have a complete recrystallization structure (Fig. 5), and the microstructure does not change with aging [60]. Therefore, dislocation strengthening is not considered in this study.

4.2.1. Precipitation strengthening

Various state variables were used to determine precipitation strengthening. Using precipitation as the interaction between the line defect and the point defect, the equation can be expressed as [55].

$$\sigma_p = \frac{M\bar{F}}{bL} \quad (6)$$

where M represents the Taylor factor, which is considered to be 3.06 [55]. $\bar{F}$, b , and L represent the average force between the precipitate and dislocation, the Burgers vector, which is 0.284 nm [55], and the effective average spacing between precipitates along the slip plane, respectively. When L is expressed with the volume fraction f and the equivalent radius of the precipitate cross-section r , Equation (6) becomes [56].

$$\sigma_p = \frac{M}{br} (2\beta G b^2)^{-1/2} \left(\frac{3f}{2\pi} \right)^{1/2} \bar{F}^{3/2} \quad (7)$$

and r could be written as [26]:

$$r = \sqrt{\frac{A}{\pi}} \quad (8)$$

where A represents the average cross-sectional area of the precipitates (Table 3). The difference in obstacle strength is reflected in the increase in YS values. The precipitation equivalent radius in this study was less than the reported critical particle radius r_c , which was estimated to be 3.0 nm based on the data in Table 3 [55]. Therefore, the precipitates in this study are considered to be shearable obstacles [26,61]. The obstacle strength can then be approximated using [56]:

$$\bar{F} = 2\beta G b^2 \left(\frac{r}{r_c} \right) \quad (9)$$

where β and G represent a constant estimated to be 0.36 [56] and the shear modulus of aluminum, which is approximately 27 GPa for aluminum alloys [26,55,56]. F is proportional to r in an alloy with shearable particles.

4.2.2. Solid solution strengthening

In age-hardened aluminum alloys, solute atoms cause significant solid-solution strengthening. The contribution of solute atoms to YS σ_{ss} is proportional to the solute atom concentration C . Assuming that the contribution of each element is additive, the solid solution strengthening

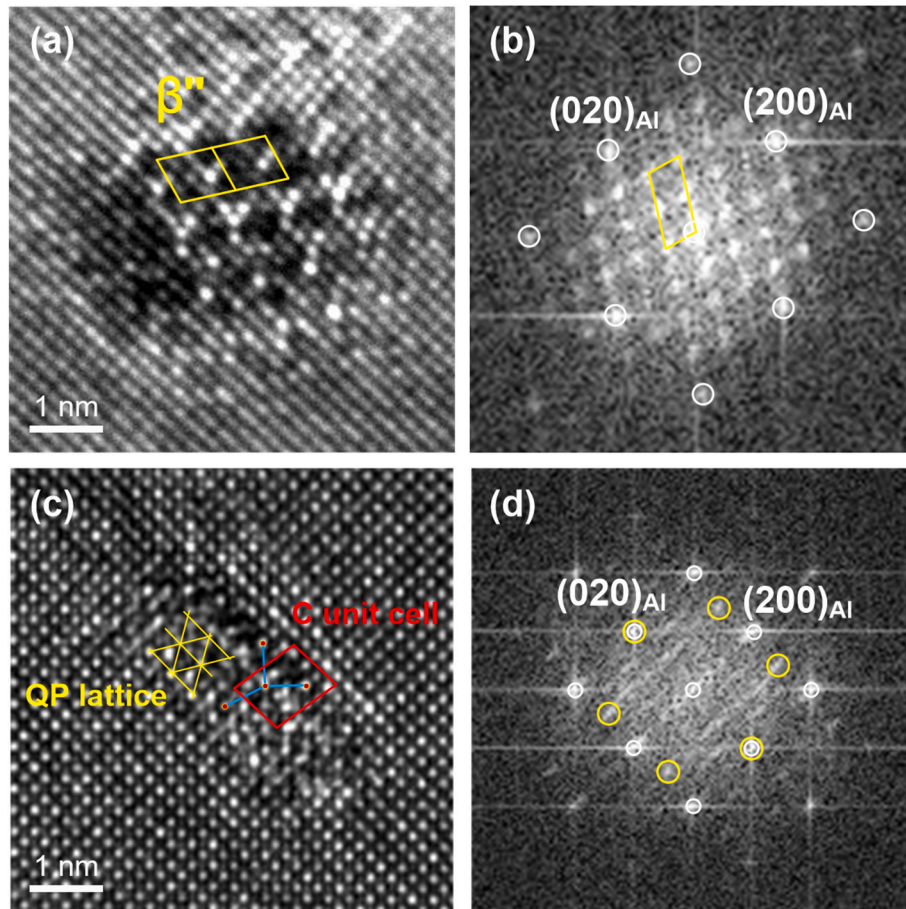


Fig. 9. HRTEM images of the two typical precipitates: (a) β'' phase, (c) disordered Q'-precursor, (b) and (d) are corresponding FFT patterns of (a) and (c).

Table 3

Quantitative calculation of the precipitates for the PA alloys.

Alloy	Average length, L (nm)	Average cross-sectional area, A (nm ²)	Average radius, r (nm)	Average number density ($10^{22}/\text{m}^{-3}$)	Volume fraction, f
0Cu	16.0 ± 6.4	8.1 ± 3.5	1.6 ± 0.3	5.3 ± 0.4	6.83×10^{-3}
0.8Cu	17.3 ± 11.8	7.0 ± 3.9	1.5 ± 0.4	6.4 ± 0.5	7.79×10^{-3}
1.2Cu	20.7 ± 13.9	5.6 ± 2.2	1.3 ± 0.3	7.5 ± 0.5	8.66×10^{-3}
1.6Cu	20.8 ± 15.1	6.4 ± 2.5	1.4 ± 0.3	7.5 ± 0.7	9.97×10^{-3}

can be expressed as [55,56].

$$\sigma_{SS} = \sum_i k_i C_i^{\frac{2}{3}} \quad (10)$$

where k_i represents the scale coefficient of element i , that is, $k_{Mg} = 29 \text{ MPa (wt.\%)}^{-2/3}$ [56], $k_{Si} = 66.3 \text{ MPa (wt.\%)}^{-2/3}$ [56], $k_{Cu} = 46.4 \text{ MPa (wt.\%)}^{-2/3}$ [56], and C_i represents the mass fraction of element i .

This study uses the method of Wang et al. [62] to measure the concentration of solute atoms and proposes an approximation. To achieve the maximum theoretical contribution of solid-solution strengthening, it is assumed that following solution treatment, all Mg and Si atoms in the alloy are dissolved in the Al matrix, except those in the Fe-bearing phases. The following are the specific parameters.

- (1) Under PA conditions, 60% of the solute atoms form precipitates [26,62].
- (2) Cu atoms replace part of the Si atoms in the precipitation sequence of the Al–Mg–Si–Cu alloy, converting the β'' phases into Cu-containing phases [63]. In the remainder of this chapter, these Cu-containing phases (including the Cu-containing Guinier–Preston (GP) zones and the precursors of Q' phase) are expressed as p-Q', and with the unconverted β'' is assumed to be 30% [28, 63].
- (3) Because the Cu matrix concentration decreases rapidly during AA due to the formation of Cu-containing precipitates and the segregation of Cu atoms at the precipitate/matrix interface [64, 65], the solution-strengthening contribution of Cu in a PA state is estimated to be 30% [66]. The consumed Cu atoms did not contribute to the solid-solution strengthening.

It should be noted that the change in solubility due to temperature was ignored in this case. To calculate the concentration C_i , the number of solute atoms consumed by the aging precipitation was subtracted from the total number of solute atoms dissolved in the Al matrix. Therefore, the concentrations of each element (C_{Mg} , C_{Si} and C_{Cu}) was determined as follows:

$$C_{Mg} = 0.4 C_{Mg}^{Total} \quad (11)$$

$$C_{Si} = C_{Si}^{Total} - C_{Si}^{Fe-bearing phase} - (C_{Si}^{\beta''} + C_{Si}^{p-Q'}) \quad (12)$$

$$C_{Cu} = 0.3 C_{Cu}^{Total} \quad (13)$$

where $C_{Si}^{Fe-bearing phase}$ can be determined using the thermodynamic re-

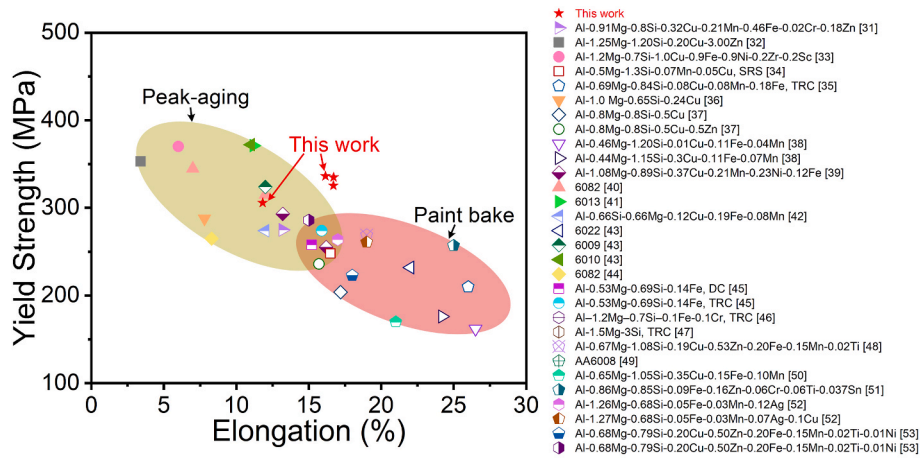


Fig. 10. Yield strength and elongation combinations of 6xxx alloys in peak-aging and paint bake state.

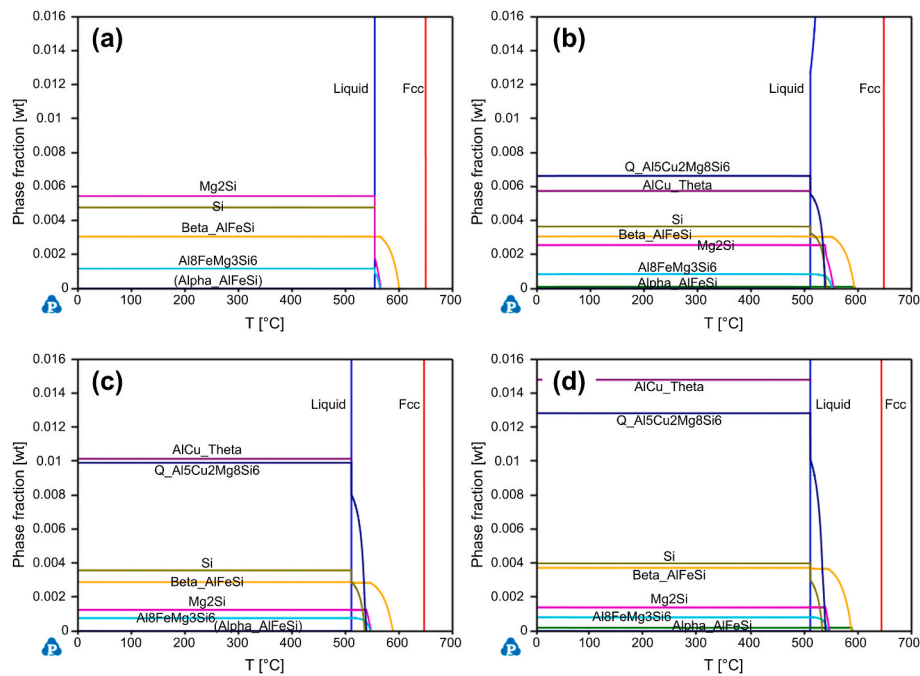


Fig. 11. Phase fraction simulated by Scheil model using Pandat software: (a) 0Cu alloy, (b) 0.8Cu alloy, (c) 1.2Cu alloy, and (d) 1.6Cu alloy.

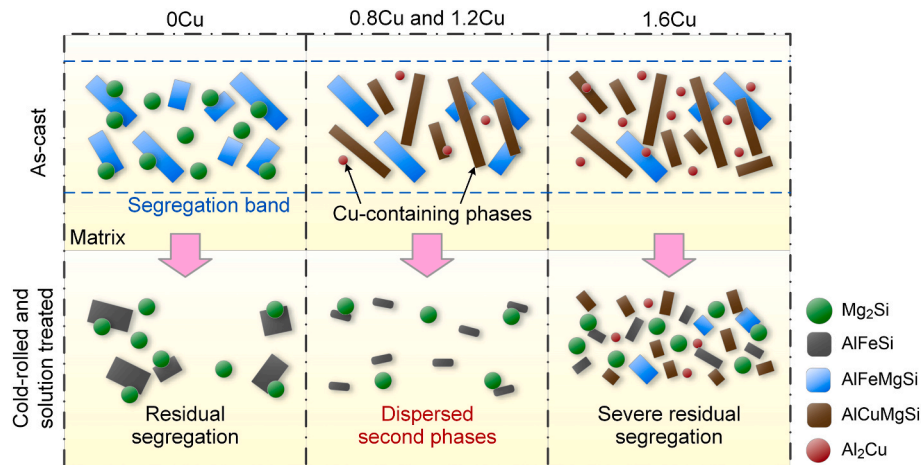


Fig. 12. Schematic illustration of the central segregation evolution process.

sults demonstrated in Section 4.1. $C_{Si}^{p''}$ and $C_{Si}^{p-Q'}$ are the element concentrations consumed by precipitation, which can be calculated using the mass relationship. Table 4 summarizes the calculated solute contents, and Table 5 lists the calculated strength values. In Section 3.3, σ_{exp} denotes the experimental YS.

The YS value of the 1.6Cu alloy was about 53 MPa lower than the calculated value, as shown in Table 5, which could be due to severe residual central segregation following cold rolling and solution treatment (as shown in Fig. 4d). The high concentration of solute elements near the segregation band results in lower supersaturation of solute atoms in 1.6Cu alloys compared to 0.8Cu and 1.2Cu alloys, which is detrimental to solution-strengthening contribution and precipitate formation. These results show that Cu is an important element with significant benefits for TRC Al–Mg–Si alloys. It has been verified that Cu can effectively modify the central-segregation composition of TRC Al–Mg–Si alloys and reduce the difficulty of dissolving the segregation during heat treatment. However, the addition of Cu up to 1.6 wt% can lead to reverse effects. In general, the addition of Cu to TRC Al–Mg–Si alloys not only modified the central-segregation composition but also improved the mechanical properties. The Cu content, especially, should be carefully chosen for the fabrication of TRC Al–Mg–Si alloy.

5. Conclusions

In this study, Al–Mg–Si strips with varying Cu contents were produced using TRC with SRS. The effects of Cu on the segregation, mechanical properties, and precipitation were evaluated synthetically, yielding the following results.

- (1) The addition of Cu promoted the segregation-phase transformation from Mg_2Si to Cu-containing phases (Q -AlCuMgSi and θ -Al₂Cu). In this study, alloying with Cu did not alleviate the as-cast segregation, but the central segregation was almost dissolved in 0.8Cu and 1.2Cu alloys after cold rolling and solution treatment.
- (2) The segregation-phase transformation from the high-melting-temperature Mg_2Si to the low-melting-temperature Cu-containing phases (~ 565 – 520 °C) reduced the difficulty to dissolve the segregation during heat treatment. Therefore, the mechanical properties of TRC Al–Mg–Si alloys were improved. Especially, the addition of 1.2 wt% Cu increased YS from ~ 306 to 335 MPa, UTS from ~ 349 to 396 MPa, and El from ~ 13 to 17% at peak aging state.
- (3) The dissolution of central segregation in 0.8Cu and 1.2Cu alloys was advantageous for the subsequent precipitation and solution strengthening. Under PA conditions, quantitative TEM analysis confirmed that adding Cu to TRC Al–Mg–Si alloys resulted in finer precipitates with a higher number density. The finer precipitates and improved solution strengthening contributed to the higher mechanical properties in Cu-containing alloys.

CRedit authorship contribution statement

Xian Wu: Validation, Data curation, Writing – original draft. **Zhi-Ping Guan:** Investigation, Conceptualization, Supervision. **Zhi-Zheng Yang:** Formal analysis, Visualization. **Xuan Wang:** Data curation, Methodology, Writing – review & editing. **Feng Qiu:** Validation, Supervision, Writing – review & editing. **Hui-Yuan Wang:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 4

Solute elements contents related to solid solution strengthening contribution (wt.%).

Alloy	C_{Mg}	$C_{Si}^{Fe-bearing phase}$	$C_{Si}^{p''}$	$C_{Si}^{p-Q'}$	C_{Si}	C_{Cu}
0Cu	0.344	0.074	0.715	–	0.301	–
0.8Cu	0.340	0.065	0.212	0.321	0.412	0.231
1.2Cu	0.340	0.059	0.212	0.321	0.448	0.366
1.6Cu	0.396	0.073	0.247	0.374	0.516	0.495

Table 5

Calculations of yield strength (MPa).

Alloy	σ_{gs}	σ_p	σ_{ss}	σ_y	σ_{exp}
0Cu	39.3	232.6	44.0	315.9	305.9
0.8Cu	37.4	238.8	68.4	344.6	325.4
1.2Cu	35.0	239.3	76.7	351.1	334.9
1.6Cu	36.9	265.4	87.3	389.6	336.3

Data availability

Data will be made available on request.

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